

# JGR Solid Earth

## RESEARCH ARTICLE

10.1029/2022JB024775

### Key Points:

- We introduce a new learning approach called neural ordinary differential equation differential learning for time series prediction
- We present the application of the method for polar motion time series prediction, focusing on the linear form of the method
- Rigorous tests show that this method can achieve state-of-the-art accuracy for both short and long-term predictions

### Supporting Information:

Supporting Information may be found in the online version of this article.

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### Citation:

Kiani Shahvand, M., Schartner, M., & Soja, B. (2022). Neural ODE differential learning and its application in polar motion prediction. *Journal of Geophysical Research: Solid Earth*, 127, e2022JB024775. <https://doi.org/10.1029/2022JB024775>

Received 13 MAY 2022

Accepted 3 NOV 2022

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## Neural ODE Differential Learning and Its Application in Polar Motion Prediction

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**Abstract** This paper introduces a new learning algorithm for accurate, physically driven time series prediction. The fundamental assumption behind the method is that the phenomena follow Ordinary Differential Equations. We investigate the general case where the time series follows an ODE of degree  $m \in \mathbb{N}$ . The resulting method is a learning algorithm based on the finite differences between the values of time series. We present the application of the method in the field of geodesy for polar motion prediction, the main objective of the present paper. We show that in this application, the linear form of the method is sufficient and offers competitive predictive performance. We present a baseline solution, in which we use historical polar motion time series from 1976 to predict up to the year 2020. The prediction horizon in this case is short-term (up to 10 days into the future). In addition, we compare the prediction accuracy in the short-term horizon with the results of the best performing model in the first Earth Orientation Prediction Comparison Campaign. On average, a 53% improvement in prediction performance is achieved. In further analyses, we compare the prediction accuracy for both short-term and long-term against the results of state-of-the-art methods, namely Multichannel Singular Spectrum Analysis, and a combination of Singular Spectrum Analysis and Copula sampling. We show that the proposed method in this paper can outperform the mentioned two methods in both short and long-term horizons, with an average improvement of the prediction performance of 54% and 52%, respectively.

**Plain Language Summary** Physical phenomena are usually represented in mathematical terms by (ordinary) differential equations. Polar motion is an important physical parameter that describes the changes in the Earth's rotation axis with respect to the surface of the Earth. There is, however, a latency of up to several days by which the values are provided by operational data centers. Therefore, predicted values are used for applications such as spacecraft navigation that would need instantaneous information about polar motion. In order to predict polar motion time series, we introduce a new learning approach that is based on fitting ordinary differential equations to data. We specifically focus on second-order differential equations, since they are typically more accurate and many physical phenomena follow an equation of this type. We demonstrate by extensive comparative analyses that our proposed method outperforms other methods both in short and long-term predictions.

## 1. Introduction

Polar motion components, denoted usually by  $x_p$  and  $y_p$ , describe the position of the Earth's instantaneous rotation axis with respect to the Earth's surface (Gross, 2007). They are usually provided as coordinates in a left-handed coordinate system, the third axis of which coincides with that of the geocentric one on the first day of the year 2000 (Altamimi & Dick, 2020). As the accurate determination of polar motion parameters requires post-processing of data collected by space geodetic techniques across the globe (Noll, 2010; Zajdel et al., 2021), there is a latency of up to several days by which the values of these parameters are provided with the highest accuracy. However, some applications such as satellite navigation, spacecraft tracking, and orientation of deep-space telescopes require instantaneous information about the Earth Orientation Parameters (EOPs) including polar motion components (Modiri et al., 2020). Therefore, it is necessary to have accurate predictions of these parameters. As such, polar motion prediction is an important task in the field of geodesy (Modiri et al., 2018).

There have been several studies on the subject of polar motion prediction, including the use of a harmonic model and autoregression (Kalarus et al., 2010), artificial neural networks (Kucak et al., 2017), a combination of Singular Spectrum Analysis (SSA) with Copula sampling (Modiri et al., 2018), and Multichannel Singular Spectrum Analysis (MSSA, Jin et al. [2021]).

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The harmonic model combined with autoregression is based on fitting sine and cosine functions with certain periods, usually yearly (365.25 days) and Chandler's 433 days, to the polar motion time series and then extrapolating this functional model at the desired prediction times using autoregression. The harmonic model is in fact the current default method in the predictions of polar motion components by the International Earth Rotation and Reference Systems Service (IERS, see Bulletin A from <https://www.iers.org/IERS/EN/DataProducts/EarthOrientationData/eop.html>). In addition, during the first Earth Orientation Parameters Prediction Comparison Campaign (EOPPCC, Kalarus et al. [2010]) this model achieved the best prediction accuracy, with maximum mean absolute errors (in terms of milliarcseconds) for a 10-day prediction period of 3.32 and 1.69 for the  $x_p$  and  $y_p$  components, respectively. The advantages of this model are its simplicity and interpretability. In fact, the model performs well at intervals where the periodicities of the polar motion components are more uniform. This implies that the evaluation part follows the same periods found in the training set, which was mostly true for the prediction period of the first EOPPCC. The predictions will be less accurate, however, if the training part is not similar to the evaluation part, in a way that can be learned by this linear statistical model. Another disadvantage of the harmonic model is its unidimensionality, which means using extra features for the prediction is not possible and only the polar motion components are individually used for predictions. This prohibits the use of other useful sources of information, which may contribute to prediction accuracy. In fact, additional information on the physics driving such phenomena is ignored and the method is purely statistical.

The use of artificial neural networks in the prediction of EOPs was first investigated by Schuh et al. (2002). An advantage of neural networks is their ability to integrate different features for the purpose of prediction. As such, different relations can be found among different time steps in the training window and also across different features, which would be otherwise unknown to most linear statistical models (except for, to some extent, multi-variable statistical models, Alexopoulos [2010]). The complexity and non-linearity of these methods, however, prohibit full interpretability. In addition, because of their usually heavily-parametrized nature, these algorithms require comparatively more data for their full training than do the linear statistical methods (Adadi, 2021; Banko & Brill, 2001; Cerqueira et al., 2019). This would pose a problem in periods for which few data points are available. For longer time spans, however, the advantages of the neural networks can be clearly observed, including higher accuracy and better exploitation of different sources of physical information (Cerqueira et al., 2019; Kiani Shahvandi & Soja, 2021a, 2021b; Makridakis et al., 2018).

Prediction using singular spectrum analysis (Golyandina & Zhigljavsky, 2020) was realized by Modiri et al. (2018). In order to take advantage of other sources of information, the method was combined with the Copula sampling technique, which can be exploited for creating synthetic data based on the distribution properties of the data fed to it (Durante & Sempi, 2021). They achieved comparable and even better prediction results than those provided by IERS Bulletin A. One of the primary conclusions was that at different periods, the prediction accuracy of the method can vary significantly. Therefore, the performance of the method is strongly dependent on the time interval during which it is evaluated. For instance, in the analysis of the  $x_p$  component of polar motion, the mean absolute errors (in terms of milliarcseconds) for predictions between 1 and 365 days have a maximum of 30.46 and 14.53, for the evaluation intervals as the years 2006 and 2009, respectively.

Another method that has its roots in SSA is the MSSA model, a method for the decomposition of a time series into several series the sum of which is the original one. As singular value decomposition, the underlying principle in this method, determines the most important constituents of time series using the so-called transition matrix and its singular values (Golyandina & Zhigljavsky, 2020), it can be used to reconstruct and predict the time series using the most relevant information in data. In fact, after the decomposition, the time series could be reconstructed with the most important eigenvalues while discarding the more high-frequency, noise-like signals (irrelevant information for prediction). In the prediction phase, a linear combination is used in an autoregressive form. The model can be used for both short (up to 10 days to the future) and long-term (up to 365 days to the future) predictions of polar motion, achieving maximum prediction mean absolute errors (in terms of milliarcseconds) for predictions of up to 360 days of 22.27 and 24.65 for  $x_p$  and  $y_p$ , respectively. As an advantage of this method, one can mention the simplicity in terms of implementation. The disadvantage is, however, that the model is not as capable as other methods like neural networks in exploiting other features that would help in the prediction process. In addition, since this method is based on autoregression, in periods of low data quality the noise gets amplified and the predictions get less accurate.

As highlighted above, most state-of-the-art methods do not take full advantage of different sources of data, including Effective Angular Momentum (EAM) functions (Chin et al., 2009; Dill & Dobslaw, 2010; Dobslaw & Dill, 2018; Dobslaw et al., 2010). These are the primary physical variables that correspond to polar motion. In general, EAM functions are of four types (as is the case with EAM provided by GFZ German Research Center for Geosciences): (a) Atmospheric Angular Momentum (AAM); (b) Hydrological Angular Momentum (HAM); (c) Oceanic Angular Momentum (OAM); (d) Sea-Level Angular Momentum (SLAM). They are of benefits in the prediction of EOPs (Dill & Dobslaw, 2017), including  $dUT1 = (UT1 - UTC)$  and Length of Day (LOD, Kalarus et al. [2010]). In addition, it is revealed that they can be closely related to the changes in polar motion components. Therefore, methods that can incorporate different features in addition to the main variables under consideration are of particular interest. In practice, the inclusion of the EAM function can provide the best prediction performance, as demonstrated by the use of AAM in the prediction of dUT1 and LOD based on Kalman filtering during the first EOPPCC. Most methods, however, only consider the effect of AAM and not other types of EAM. It could be advantageous to use all types of EAM functions, as especially it is the OAM that would have the most important effect on the prediction of polar motion (Dobslaw et al., 2010). Hence, we focus on the methods that can use various features. For this purpose, we pay close attention to machine learning methods.

A characteristic of a wide range of physical phenomena is that they follow differential equations (Lein, 2021), often of the ordinary type. However, the exact form of the equation may not be known, or may not be accurate enough. In these cases, finding the function of the differential equation is of primary interest. This can be done by fitting a stochastic function to data such that it satisfies the differential equation. In the case of differential equations of the ordinary type, the so-called method of Neural Ordinary Differential Equations (Neural ODEs, Chen et al. [2018]) can be applied. Neural ODEs are gaining increasing attention because of their applications in many different fields, including computer science (Tuor et al., 2020), physics (Liang et al., 2021), and robotics (Meleshkova et al., 2021). The advantages of this approach include the use of highly accurate neural networks and the possibility to include physical information. As polar motion components can be considered to follow Ordinary Differential Equations (ODEs, Chin et al. [2004]; Gross [2007]), the Neural ODEs method can be applied.

The objectives of the present study can therefore be summarized as follows:

- Development of a learning method based on Neural ODEs, with a special focus on polar motion prediction, which is capable of handling an arbitrary number of features
- Inclusion and analysis of the effects of the EAM functions including AAM, HAM, OAM, and SLAM

The subsequent parts of this study are organized as follows: In Section 2, the formulas of the method are presented. In Section 3, the data used in the study is described. In Section 4, the application of the method is demonstrated. Section 5 deals with the analysis of the results. Finally, Section 6 is devoted to conclusions. The proofs of the theoretical results and further accompanying information are presented in the Supplementary Materials.

## 2. Methodology

In this section we present the mathematical formulas for the differential learning approach. The proofs can be found in the Section A.1–B in Supporting Information S1. As highlighted above, the idea is to consider the time series values following an ODE of degree  $m \in \mathbb{N}$ . In the following section, the problem is formulated and the solution is presented.

### 2.1. Introducing Differential Learning Method

Denoting the time series under consideration as  $y(t)$  – which is considered a function of time only – the differential learning problem is defined as finding the (stochastic) ODE function  $f$  (Kotelenez, 2008) that satisfies the following relation:

$$\frac{d^m y}{dt^m} = f(y, y^{(1)}, \dots, y^{(m-1)}, t) \quad (1)$$

in which  $y^{(i)}$ ,  $i = 1, \dots, m - 1$  are the derivatives of  $y$ . The purpose of the algorithm is to solve this equation in its general form. Note that the values of  $y$  at different time steps (i.e.,  $y(t_1)$ ,  $y(t_2)$ , ...) are denoted by  $y_1$ ,  $y_2$ , ...

**Remark 1.** Note that in order to solve Equation 1 for a stochastic function  $f$  we assume that the function is only a function of previous values of  $y, y^{(1)}, \dots, y^{(m-1)}$ . We, therefore, eliminate time as an explicit variable, but it is kept implicit in the preceding values of  $y$  (This can be justified by the fact that variables do not have a direct dependency on time, but rather indirectly via features at different times). For now, we assume a regularly sampled time series. However, in Section 2.2 a differential learning method suitable for irregularly-sampled (also called unevenly-spaced) time series (Erdogan et al., 2005) is discussed.

The solution of Equation 1, at time step  $n$  can be represented as follows:

$$y_n = y_{n-1} + \sum_{j=1}^M \omega_j F(y_{n-j}, y_{n-j}^{(1)}, \dots, y_{n-j}^{(m-1)}) \quad (2)$$

where  $M$  is the window length for the historical data (used for making the prediction),  $\omega_j, j = 1, \dots, M$  are the weights that must be determined through the process of learning, and  $F$  is the neural network that approximates  $m$ th indefinite integration of function  $f$  in Equation 1. Therefore,  $F$  is a (stochastic) function that has learnable parameters.

**Remark 2.** As one can see from Equation 2, the value at time step  $n$  is obtained from the previous time step  $n - 1$  plus a residual term that can be learned by a neural network. Therefore, we call this method differential learning. It is important to note that since according to Equation 2 the inputs to the ODE function  $F$  (being neural networks) are the previous values of the time series, the so-called Recurrent Neural Network (RNN, Rumelhart et al. [1986]) are needed. Note also that the choice of the ODE function  $F$  depends on a number of factors, including the volume and complexity of data. Another argument for using RNN models, including Elman and Long Short-Term Memory (LSTM, Hochreiter & Schmidhuber [1997]), is that they have proven to be accurate in modeling sequential data like time series. For instance, Benitez et al. (2021) showed, in a large empirical investigation, that the LSTM model is the leading machine learning algorithm for time series prediction. On the other hand, we will also use Elman, which is the simplest form of RNNs to analyze the appropriate choice of the neural network for the ODE function. Therefore, we use these models for the application presented in this study.

Based on the explanation above, if we use RNN cells in the model, we can write

$$y_n = y_{n-1} + \sum_{j=1}^M \omega_j \text{RNN}(y_{n-j}, y_{n-j}^{(1)}, \dots, y_{n-j}^{(m-1)}) \quad (3)$$

According to Equation 2, the solution to the second-order differential learning approach ( $m = 2$ ) reads

$$y_n = y_{n-1} + \sum_{j=1}^M \omega_j F(y_{n-j}, y_{n-j}^{(1)}) \quad (4)$$

Linear differential learning is defined based on the choice of linear function for  $F$  in Equation 4, having the solution

$$y_n = \sum_{j=1}^M \alpha_j y_{n-j} + \beta_j y_{n-j}^{(1)} \quad (5)$$

in which  $(\alpha_j, \beta_j), j = 1, \dots, M$  are learnable weights.

**Remark 3.** Note that in order to derive the learnable weights we can use uncertainty in the observation files to better fit the models to the data. In case of the linear models, this would result in so-called weighted least squares (Magee, 1998). Further information is given in the proofs in Supplementary materials. In the case of non-linear models, we use the method proposed by Kiani Shahvandi and Soja (2022).

In case additional features (other than the value under consideration) are to be used in the method (here in the linear form), we can use the following relations, in which extra features are denoted as  $X^k, k = 1, \dots, p$

$$y_n = \sum_{j=1}^M \alpha_j y_{n-j} + \beta_j y_{n-j}^{(1)} + \sum_{j=1}^M \sum_{k=1}^p \gamma_{j,k} X_{n-j}^k \quad (6)$$

where  $\gamma_{j,k}$ ,  $j = 1, \dots, M$ ,  $k = 1, \dots, p$  are the weights that should be determined using the linear least squares problem.

In the case of non-linear models, including RNNs, one can extend Equation 2 to

$$y_n = y_{n-1} + \sum_{j=1}^M \sum_{k=1}^p \omega_j F(y_{n-j}, y_{n-j}^{(1)}, \dots, y_{n-j}^{(m-1)}, X_{n-j}^k) \quad (7)$$

The relation given in Equation 7 is the general form of the differential learning model, in which an arbitrary number of features could be used to augment the model for better prediction. In order to augment the model further, we can combine it with neural Latent ODEs, in which case we can deal with missing data or make interpolations in the hidden space. This is done in Section 2.2.

## 2.2. Combination With Method of Latent ODEs

In case we intend to handle missing values in data or include a form of simultaneous interpolation-approximation for better predictions (Rubanova et al., 2019), we can use the method of Latent ODEs in the case of using RNNs. Latent ODEs are defined in the so-called Latent space (Ali & vanKaick, 2021). This space is formed with hidden layers that contain artificial neurons. Each hidden layer is formed by applying a linear combination to the features of the preceding hidden layer and subsequently applying an activation function. Therefore, we can propose the appropriate relations as follows:

The simultaneous interpolation-approximation method for the differential learning model reads as follows:

$$y_n = y_{n-1} + \sum_{j=1}^M \sum_{k=1}^p \omega_j \text{RNN}(h_{n-j}, y_{n-j}, y_{n-j}^{(1)}, \dots, y_{n-j}^{(m-1)}, X_{n-j}^k) \quad (8)$$

where the hidden state of the RNN cell  $h_{n-j}$  is determined through the fitting of a Neural ODE to the values of hidden states, which is called Latent space (and hence the name Latent ODEs)

$$h_{n-j} = G(h_{n-j-l}, t_{n-j-l}), \quad l = 1, \dots, n-j+1 \quad (9)$$

where  $G$  is a neural network similar to  $F$  in Equation 7.

The linear case of Equations 8 and 9 reads as follows:

$$y_n = \sum_{j=1}^M \alpha_j y_{n-j} + \beta_j y_{n-j}^{(1)} + \eta_j h_{n-j} + \sum_{j=1}^M \sum_{k=1}^p \gamma_{j,k} X_{n-j}^k \quad (10)$$

where

$$h_{n-j} = \sum_{l=1}^{n-j+1} C_l h_{n-j-l} \quad (11)$$

in which  $C_l$  are weights that are learned jointly with  $(\alpha_j, \beta_j, \eta_j)$ .

In order to analyze the importance of different features, it is needed to use the concept of sensitivity analysis. This has been done in the following section.

## 2.3. Global Sensivity Analysis for Feature Importance

In order to perform sensitivity analysis (Saltelli et al., 2008) for the mentioned method, the goal of which is the determination of the importance of different features, we can use the method of global sensitivity analysis. If the model is the one given in Equation 6, then we can simply use the method of variance decomposition for linear

models. Therefore, in the second-order ordinary differential equations for the determination of the Importance of Feature (IF) we have

$$\begin{aligned} IF_j^y &= \frac{\alpha_j \sigma_{yM}}{\sigma_y} \\ IF_j^{y(1)} &= \frac{\beta_j \sigma_{yM}^{(1)}}{\sigma_y} \\ IF_{j,k}^{X^k} &= \frac{\gamma_{j,k} \sigma_{X_M^k}}{\sigma_y} \end{aligned} \quad (12)$$

where  $\sigma$  denotes the standard deviation and the subscript  $M$  means the values are computed in the input window. Further theoretical results on the uncertainty propagation are given in the Supplementary materials.

## 2.4. Summary of the Methods

In the following, a summary of the methods used in this study is provided. Based on what is mentioned earlier, six different categories of the differential learning approach are defined as the following:

- Linear model: a simple recursive formulation as in Equation 5
- Linear model + latent ODE method: simultaneous interpolation and approximation, given in Equations 10 and 11
- Non-linear LSTM model: a model based on recurrent cells when the RNN cell in Equation 8 is the LSTM layer
- Non-linear LSTM model + latent ODE method: a model the same as Equations 8 and 9 when the RNN cells are of the LSTM type
- Non-linear Elman model: a model based on recurrent cells when the RNN cell in Equation 8 is the Elman layer
- Non-linear Elman model + latent ODE method: a model the same as Equations 8 and 9 when the RNN cells are of the Elman type

The linear model is used mostly when the time series is smooth. The non-linear model is suitable for more variable time series. As mentioned, examples of non-linear models include LSTM and Elman recurrent neural networks. However, it is necessary to have significantly more training data compared to the case of linear models. In case of missing data or in case of utilizing a relatively short time series, a combination of differential learning with Latent ODEs is best suited. The structure of non-linear models used in the studies is presented in the Section C in Supporting Information S1.

## 3. Data Description

In this section, we describe the data used for four different analyses for polar motion predictions in the next sections. These four studies are as follows:

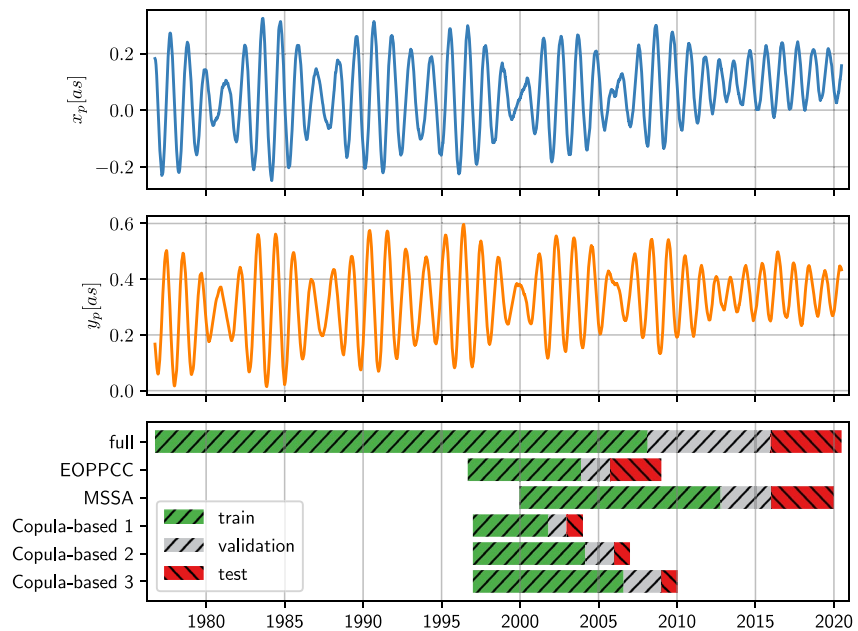
1. baseline solution: utilizing a full EOP dataset up to the year 2020 (Section 4.1)
2. under the first EOPPC conditions (Section 4.2)
3. under the conditions of a study by Jin et al. (2021) to compare our methods with predictions of the MSSA-based method (Section 4.3)
4. under the conditions of a study by Modiri et al. (2018) to compare our methods with predictions of the Copula-based method (Section 4.4)

Figure 1 depicts the polar motion time series as well as the training and prediction time spans used within the individual analyses.

The data sets used for these studies belong to two categories: EOPs time series and EAM data.

As per the requirements of the proposed method, in addition to the values of polar motion, it is necessary to have the rates of each of the components as well. Since the rates are not available in the IERS products, the EOP solutions provided by Jet Propulsion Laboratory (JPL, <https://keof.jpl.nasa.gov/combinations/2019/>) are used. These data files include the polar motion values and their corresponding rate of change, as well as other EOPs such as LOD and dUT1 values (Ratcliff & Gross, 2020). The values are provided with daily resolution at 0 UTC from 28

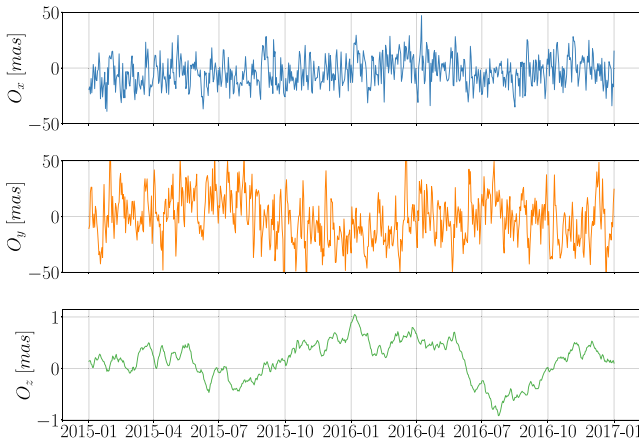




**Figure 1.** Top: polar motion  $x_p$ . Center: polar motion  $y_p$ . Bottom: time spans used for training, validation, and testing for individual comparisons. The first study (full) is the baseline we present. The second study is the comparison with the best-performing method of the first Earth Orientation Prediction Comparison Campaign. The third study refers to the comparison with the predictions of the Multichannel Singular Spectrum Analysis-based method. The three Copula studies refer to the three prediction intervals 2003, 2006, and 2009.

September 1976–26 June 2020, determined through the combination of several space-geodetic techniques such as the Global Navigation Satellite System (Zajdel et al., 2021) and geodetic Very Long Baseline Interferometry (Sovers et al., 1998). It is important to note that besides polar motion, other EOPs such as dUT1 and LOD are also used as features for the prediction of polar motion. However, for self-evaluation purposes of our polar motion predictions, as done in Section 4.1, the reference EOPs data are those provided by JPL (i.e., of the same type as training data). In the comparative analyses with the results of other methods, however, the official EOPs data provided by the IERS (<https://www.iers.org/IERS/EN/DataProducts/EarthOrientationData/eop.html>) is used as a reference. These include the analyses in Sections 4.2, 4.3, and 4.4.

In order to couple the proposed method with the physical information about the changes in Earth's rotation, EAM data are included as additional features. For the EAM data, the solution provided by the GFZ Potsdam (<http://rz-vm115.gfz-potsdam.de:8080/repository>) is used (Dill et al., 2019; Dobsław & Dill, 2018). In order to use all available physical information, all three components ( $x$ ,  $y$ ,  $z$ ) of each of the EAM types, namely AAM, HAM, OAM, and SLAM are utilized. However, it should be noted that theoretically, only ( $x$ ,  $y$ ) components are of importance for the prediction of polar motion, as stated in the study by Chin et al. (2004). We decided to include the  $z$  component, because of the possible correlation between the different features used in the input data. In fact, our tests show that removing the  $z$  component would result in approximately 1%–2% reduction in prediction performance, more importantly in nonlinear models. While AAM, HAM, and OAM include contributions from mass and motion, SLAM only contains a contribution from the mass. Besides EAM observations, also forecast files are included as features. GFZ provides two forms of forecast files, 6-day forecasts and 90-day forecasts. Previous studies by Dill and Dobsław (2017) highlighted that especially the first six forecast days are of importance. Reasons are that GFZ processes numerical weather prediction data only for 6 days into the future, which is then directly used for both AAM calculation and the forcing of ocean and land surface models that deliver OAM, HAM, and SLAM. The additionally published 90-day-long combined forecasts additionally make use of least-squares-based extrapolation and autoregressive modeling (Dill & Dobsław, 2017) to extend predictions to a forecast range typically covered by IERS EOPs products. Therefore, the 6-day forecast files are used within this study.



**Figure 2.** The components of the sum of all effective angular momentum functions-observations ( $o_x, o_y, o_z$ ) for the years 2015–2016.

It is important to note that the predictions are only available for historical data since the year 2016. Historical observations from 1 January 1976 are used to generate so-called pseudo-forecasts. This can be done using two approaches.

- Addition of random noise
- Simulation using Copula theory

The first approach is done through the addition of random Gaussian noise with standard deviation taken from the a posteriori analysis by GFZ for different days of prediction (Dobslaw & Dill, 2019). Therefore, we have the following relation:

$$f_{i,c}^t = o_c^t + n_i(0, \sigma_i), \quad i = 1, \dots, 6 \quad (13)$$

in which  $f_{i,c}^t$  is the  $i$ th pseudo-forecast of type  $t \in \{\text{AAM, HAM, OAM, SLAM}\}$  for component  $c \in \{x, y, z\}$ ;  $\{o_c^t\}$  is the corresponding observation, and  $n_i$  is the random Gaussian noise with standard deviation  $\sigma_i$ . The final observation and pseudo-forecast values used in the proposed method are based on the component-wise summation of all the EAM functions. This is done for both the observations and forecasts. In mathematical terms:

$$\begin{aligned} o_{i,c} &= \sum_t o_{i,c}^t \\ f_{i,c} &= \sum_t f_{i,c}^t \end{aligned} \quad (14)$$

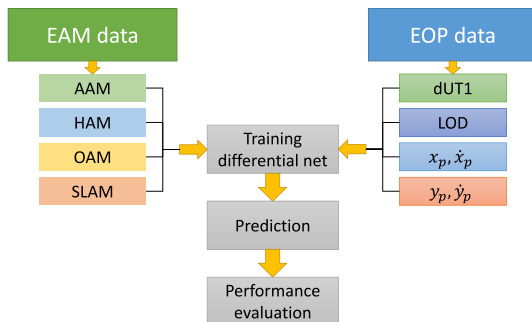
The second approach is based on copula sampling (Jaworski et al., 2010). Data from 2016 onward are used to fit a Copula model and then sampling from the fitted model. The advantage of this approach is its capability of preserving statistical properties when sampling from the model.

It can be shown that both observations and forecasts of EAM follow a Gaussian distribution. Therefore, a multi-dimensional Gaussian distribution is considered the Copula model. If we denote MGD as the multidimensional Gaussian distribution, the pseudo-forecasts are generated as follows

$$f_{i,c}^t = o_c^t + \text{sample}(\text{MGD}_i), \quad i = 1, \dots, 6 \quad (15)$$

**Remark 4.** Unless otherwise stated, the default method for the generation of pseudo-forecasts in the analyses in the results section is Copula sampling. In addition, it is important to note that the prediction of  $x_p$  and  $y_p$  are made individually, meaning separate models are trained and evaluated for each of these two components.

In this manner, the algorithm focuses on one target and thus achieves better accuracy compared to the case where they are predicted simultaneously. In our tests, this has been verified.

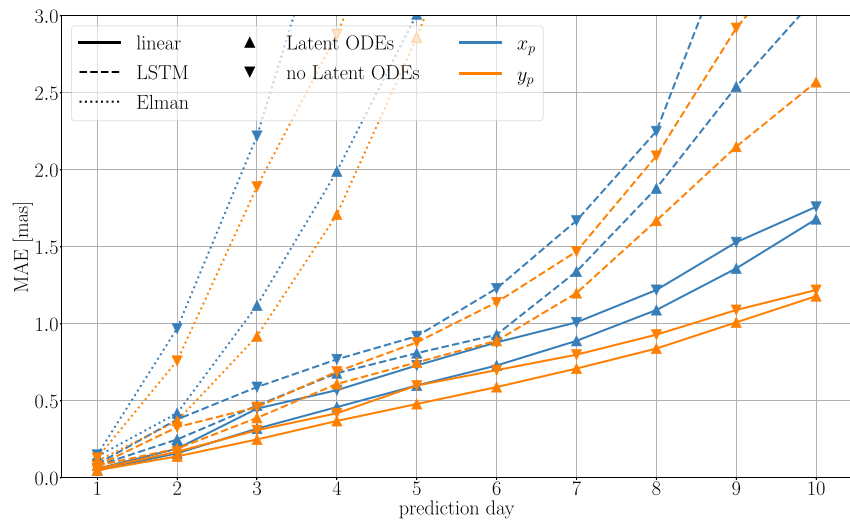


**Figure 3.** The suggested algorithm for the prediction and evaluation of polar motion. Earth orientation parameters data (dUT1, length of day, and polar motion components  $x_p, y_p$  together with their rates  $\dot{x}_p, \dot{y}_p$ ) are used in combination with effective angular momentum data (atmospheric angular momentum, hydrological angular momentum, oceanic angular momentum, sea level angular momentum) to predict polar motion. Note that for  $x_p$  and  $y_p$  separate models are trained and evaluated individually.

Figure 2 depicts the EAM observations for the years 2015 and 2016. The similar figure for individual EAM components, that is, AAM, HAM, OAM, SLAM, is presented in Figure S2 in Supporting Information S1. Note that the values are converted to milliarcseconds by multiplying the original, dimensionless values with 86,400,000 (This coefficient is derived by multiplying the number of seconds in a day [86,400] by 1000). The figure shows the three dimensional coordinates of EAM, in which the  $z$  component is dominated by the  $z$  component of AAM.  $x$  and  $y$  components have relative contributions from all the EAM functions.

The aforementioned steps of the analyses are summarized in Figure 3. The EAM and EOPs data are used to train the models, which will then be used for prediction purposes. Conclusions are then drawn based on how well the models perform on the test sets.





**Figure 4.** Prediction accuracy of the six different methods as mentioned in Section 2.4. Methods in which Latent ordinary differential equations (ODEs) are used are shown by the marker  $\Delta$ , while those without Latent ODEs are shown by the marker  $\nabla$ .

## 4. Results and Discussion

### 4.1. Baseline Solution: Utilizing a Full EOP Dataset up to the Year 2020

In this section, we present a new baseline of our own. The properties of the analysis are as follows:

- prediction horizon of 10 days (short-term)
- training interval from 28 September 1976 to the end of 2015 (without retraining)
- prediction interval between 1 January 2016 and 26 June 2020, as depicted in Figure 1
- evaluation based on the JPL EOPs data

#### 4.1.1. Search for the Best Variation of the Proposed Method and Input Sequence Length

First, an analysis regarding the optimal input sequence length  $M$  is performed. As mentioned before, in this study we use the method of Copula sampling to generate pseudo-forecasts. Therefore, from the available training time span, only 80% is used for training while the last 20% serve as a validation set to identify the best window length  $M$ , as shown in Figure 1. For this purpose,  $M$  is changed in the set of values  $\{2, 3, 4, \dots, 60\}$  to find the best value for every six variations of the differential learning approach. In addition, the best variation of the differential learning model is chosen.

Analyses of the different variations of the proposed method in Section 2.4 result in different values of  $M$  for the best input sequence length. Combination with Latent ODEs does not change the optimal input sequence length. The values  $M = 10$ ,  $M = 30$ , and  $M = 20$  are derived for the linear, Elman, and LSTM models, respectively. Figure 4 shows the prediction accuracy of different methods across prediction days.

$$\text{MAE}_i = \frac{1}{N} \sum_{j=1}^N |O_{i,j} - P_{i,j}|, \quad i = 1, \dots, h \quad (16)$$

In Equation 16, the indices  $i, j$  refer to the  $i$ th prediction day and  $j$ th value (we have  $N$  values, which we average over).  $||$  represents the absolute value and  $O, P$  refer to observation and prediction, respectively. In this case  $h = 10$ . Further discussion on the choice of best differential learning variation and its input sequence length is presented in the Supplementary Materials.

Based on the analysis in Figure 4, the linear method when combined with Latent ODEs shows the best performance. However, the two LSTM-based methods are also performing well. The Latent ODE methods are performing better compared to their counterparts. The difference between the linear model and LSTM (both combined

**Table 1**  
Number of Samples Used in This Study, That Is, the Baseline Solution

| Set        | Number of samples |
|------------|-------------------|
| Training   | 11,488            |
| Validation | 2,862             |
| Testing    | 1,649             |

with Latent ODEs) is almost a factor of two. This implies that using the linear method is sufficient and results in the best overall accuracy. Note that we showed once all results and in the following sections only the linear one.

**Remark 5.** Since the linear model combined with Latent ODEs is performing best while simultaneously being the simplest and thus least computationally expensive method (Makridakis et al., 2018), it was selected for further evaluation.

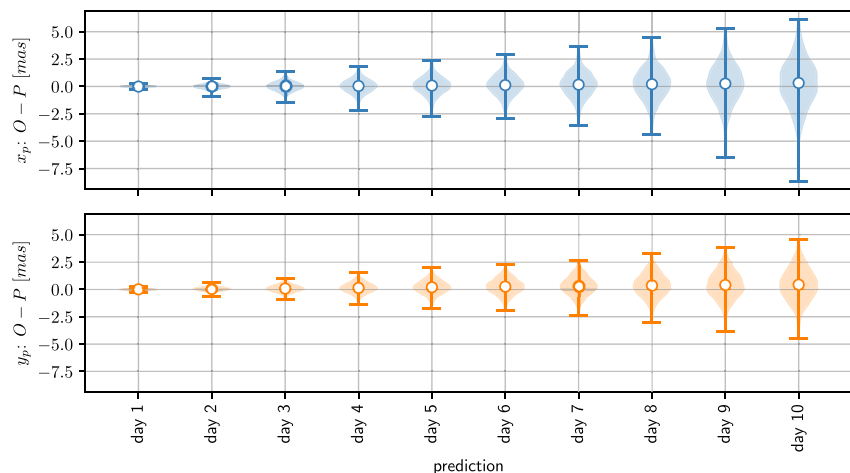
#### 4.1.2. Prediction Accuracy, Residuals, and Their Correlation

As mentioned, the best variation of the differential learning method is the linear one combined with Latent ODEs.

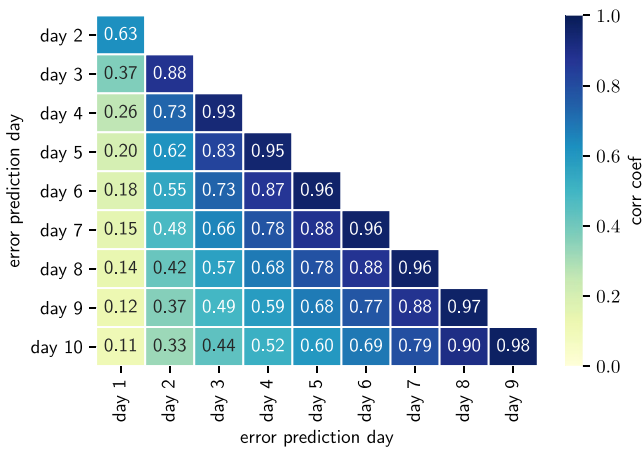
In addition to Figure 4 which depicts prediction accuracy for different methods used in this study, the numerical values for prediction accuracy for individual days are placed in Tables A1 and A2 in the Appendix. In addition, Table 1 shows the number of samples for each of the training, validation, and testing sets.

Residuals are formed when the predictions are subtracted from their corresponding observations. In Figure 5, the distribution of the residuals across different prediction days is depicted. Note that these are for the test set. The results highlighted in Figure 5 reveal that no significant bias is present in the predictions, which is important for applications relying on polar motion forecasts. The bias for the day one predictions is about  $1 \mu\text{s}$  while it grows to approximately  $30 \mu\text{s}$  for the day 10 prediction. In contrast, the range of prediction error is 1 mas for day one and grows to approximately 5 mas for day 10. Figure 6 displays their mutual correlations.

From Figures 5 and 6 one can understand that the distribution of residuals is normal Gaussian, with increasing variance (and to a lesser degree mean) values for later prediction days. The standard deviation, however, grows much more rapidly than the mean. The mean of the absolute of these residuals should theoretically correspond to the prediction performance when the evaluation sampling size tends to be very large. As per Monte-Carlo theory (Thomopoulos, 2013), with a sample size of  $N$  the convergence rate is proportional to  $\frac{1}{\sqrt{N}}$ . Here, therefore, based on the size of the testing set (Table 1) one can assume that the values that are derived can be trusted up to around 2% uncertainty. This implies that as the number of samples for training and evaluation increases, the prediction accuracy of the method is close to the values shown in Figure 4. In addition, the correlation between residuals across different days reflects the fact that the model has been sufficiently fitted to the data.



**Figure 5.** Range distribution of the residuals in the test set. The residuals in the vertical axis are displayed across different days. The white circles show the mean of the differences, while the whiskers represent the range that the predictions could take. This is equivalent to having a Gaussian distribution with increasing variance. The mean values across different days are below the uncertainty level of polar motion, starting from approximately  $1 \mu\text{s}$  for day 1 to approximately  $30 \mu\text{s}$  for day 10.



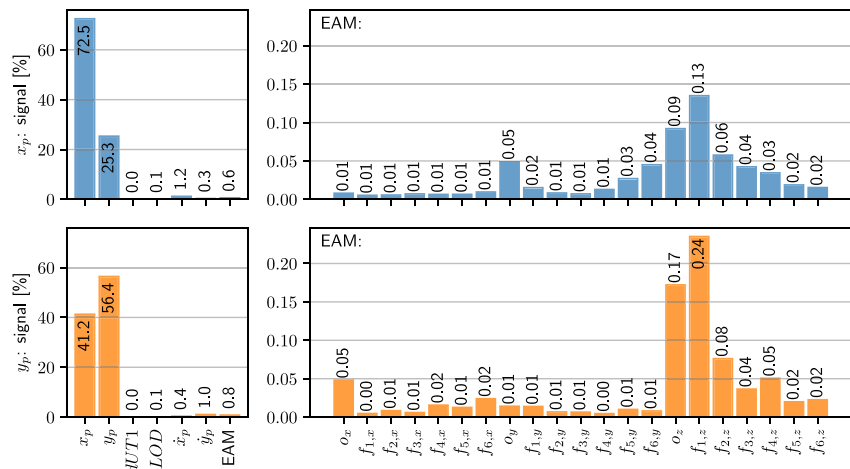
**Figure 6.** Correlation between the residuals across different days in the test set.

#### 4.1.3. Sensitivity Analysis for Different Features and Different Time Steps

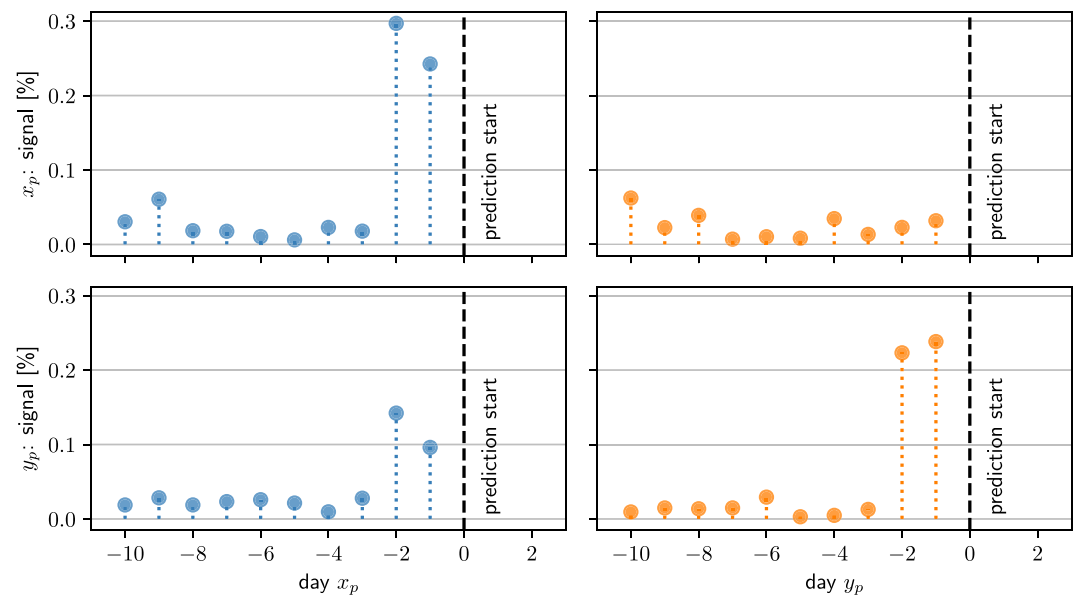
In order to quantify the contribution of each input feature in the model to the prediction, we perform a feature sensitivity analysis based on the derived formulas in Equation 12. In this case, the contribution of each feature is computed by averaging over all samples and all time steps. Similarly, an analysis is done for the importance of time steps in the input sequence. To this end, the average contribution of each feature for all samples is analyzed for each day. These are shown in Figures 7 and 8, respectively.

Considering Figures 7 and 8, it can be seen that the majority of the signal comes from the history of  $x_p$  for predicting  $x_p$  and from the history of  $y_p$  for predicting  $y_p$ . Also, it seems that the latest time steps are the most important ones, implying that the majority of the signal comes from the immediately preceding time steps. This is because the latest time steps have the highest correlation with the following values, therefore they contribute the most to the prediction, highlighting the importance of rapid processing of geodetic products. This is not surprising since the method tries to predict the polar

motion values without removing the strong periodic signals with an amplitude of approximately 200 mas, as can be seen in Figure 1. The importance of  $x_p$  for the prediction of  $y_p$  and vice versa can also be clearly seen. This is in fact related to the similarity between polar motion components ( $x_p, y_p$ ) and the processes that govern them, meaning that they change in a similar way to each other. Other important features are rates of polar motion ( $\dot{x}_p, \dot{y}_p$ ), on which the second-order differential learning method is based. In fact, using these is similar to the use of differentiation (first and higher orders) in Taylor series expansion, by which any continuously differentiable function can be approximated. As a result, the rates help approximate the changes in the function. It is important to note that these rates correspond to forward finite differentiation (LeVeque, 2007), enabling the network to predict more accurately by providing some independent information about the future value. For clarification, we consider, for instance, the case of linear differential learning. If the same analysis is done with and without using the rates, the maximum MAE would increase from 1.76 to 3.68 for the  $x_p$ , and from 1.22 to 2.97 for  $y_p$ . At a first glance, the importance of the other features including EAM seems to be insignificant since they contribute to less than 1% of the total signal. However, one has to keep in mind that the prediction accuracy that we achieve is in many cases below 1 mas as can be seen in Figure 4 and thus way below 1% for a signal with an amplitude of around 200 mas. Therefore, the inclusion of the EAM data as feature is highly relevant. For clarification: if the same predictions would be calculated without the EAM data, the MAE for different prediction days (day 1



**Figure 7.** Importance of input features to the model. The left parts show the importance of the seven features, namely,  $x_p$ ,  $y_p$ , dUT1, length of day,  $\dot{x}_p$ ,  $\dot{y}_p$ , and effective angular momentum (EAM). In the right parts, the importance of EAM is further divided into its components, that is,  $x, y, z$  and their 6-day forecasts. The upper figure is for prediction of  $x_p$ , while the lower figure is for the prediction of  $y_p$ .



**Figure 8.** Importance of different time steps in the input sequence, using linear differential learning combined with Latent ordinary differential equation. The left parts show the effect of different time steps in  $x_p$ , for both  $x_p$  (above) and  $y_p$  (below). The right parts show the effect of different time steps in  $y_p$ , for both  $x_p$  (above) and  $y_p$  (below).

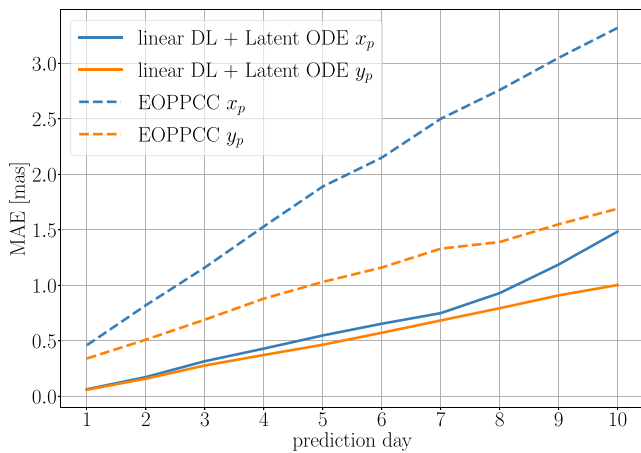
to day 10) would increase from the maximum of 1.44 (mas) to 3.13 (mas) for  $x_p$  and from 1.01 (mas) to 1.79 (mas) for  $y_p$ . Further information regarding the importance of EAM functions can be found in Section 5.1. For the EAM features, it seems like although the  $x$  and  $y$  components contribute to the prediction performance, the  $z$ -component is the most important one. Part of the reason is that since we are using polar motion rates and these correspond to the  $x$ ,  $y$  components of Geodetic Angular Momentum (GAM, Dill et al. [2020]), which is more relevant for the prediction of polar motion, using an extra feature of EAM may not be as helpful. However, the model can learn something from the  $z$  component of EAM. The reason may be the use of LOD as a feature, which is correlated with the  $z$  component of EAM. In fact, by removing the LOD and  $z$  component of EAM the prediction accuracy would drop by approximately 1%–2%. It seems, therefore, that the model is in fact learning from seemingly not-so-relevant features. We also think that the combination with Latent ODEs has also some effects, as it tries to interpolate in the hidden space, thus possibly producing some apparently minor artifacts, which have positive effects on the prediction performance. In fact, the linear model per se is less affected when the LOD and  $z$  component of EAM are removed from the input features.

#### 4.2. Analysis Under EOPPCC Conditions for Short-Term Predictions

The first EOPPCC was held from 2005 to 2008 in which more than 20 different EOP prediction methods were compared (Kalarus et al., 2010). The best method for polar motion prediction was determined to be the extrapolation of the harmonic model combined with autoregression. A simple summary of the characteristics of the comparison campaign is mentioned in the following. All of these points are also considered in this hindcast analysis:

- prediction horizon of 10 days (short-term)
- time interval for training the model between 1 September 1996 and 30 September 2005
- time interval for submitting the predictions from 1 October 2005 to 31 December 2008, every Thursday
- possibility of using various data sources, such as EOPs, EAM, etc.
- evaluation based on the IERS final EOPs data (IERS 05 C04)

This hindcast analysis specifically focuses on the short-term polar motion predictions of up to 10 days into the future. A similar analysis regarding the optimal input time length  $M$  can be performed for different variations of the differential learning method. This would again yield the same conclusions as the previous case, meaning the linear method in combination with Latent ODEs and with  $M = 10$  results in the best prediction performance. In



**Figure 9.** Comparison of the prediction performance of the best method in Earth Orientation Prediction Comparison Campaign and linear differential learning (combined with Latent ordinary differential equations) on different days in terms of MAE (mas) for both the components  $x_p$  and  $y_p$ .

terms of MAE, for  $x_p$  and  $y_p$  components the linear method in combination with Latent ODEs is at least 24.4% and 15.9% more accurate than the rest of the methods mentioned in Section 2.4.

#### 4.2.1. Comparison With the Results of Best-Performing EOPCC Method

The accuracy of the predictions up to 10 days into the future using the best-performing method from the first EOPCC (i.e., harmonic analysis combined with autoregression) and our linear differential learning method (combined with Latent ODEs) for the test interval is depicted in Figure 9 in terms of MAE.

In accordance with the results presented in Figure 9, our method outperforms the best method in the first EOPCC across all days. The same patterns in the MAEs of  $(x_p, y_p)$  can be observed, that is,  $y_p$  is predicted with better accuracy than  $x_p$ . The improvement rate of differential learning results w.r.t the first EOPCC for  $x_p$  is 86% for the first day and 55% for the 10-th day. The corresponding improvements for  $y_p$  are 82% and 40%. The average improvement across both components and all days is around 53%. As one can observe, the improvement is more evident in the case of  $x_p$ . The reason is the use of uncertainties provided for  $(x_p, y_p)$  in the observation files in the adjustment process

of deriving the coefficients of the model (see the Supplementary Materials): as the uncertainties in  $y_p$  are smaller than those of  $x_p$ , their use in the adjustment will not help as much as those of  $x_p$  will do. It is also important to note that the first day predictions show similar accuracy for both  $x_p$  and  $y_p$ . As it is pointed out (Ray et al., 2017), these accuracies are already close to the long-term accuracy of polar motion determination. The deterioration of accuracy across different days is, however, more prominent in the proposed approach. This is due to the linear nature of the used model. In addition,  $x_p$  is more affected, such that the accuracy decreases faster compared to  $y_p$ . This is again due to the bigger uncertainties in  $x_p$ .

**Remark 6.** It is worth mentioning that since the results of EOPCC were based on an operational setting, the improvement of results we presented here might be too optimistic since certain operational aspects could not be considered. These include problems with data availability (such as missing data) and using rapid EOP solutions for making predictions at the submission time (which are of lower quality compared to the final products).

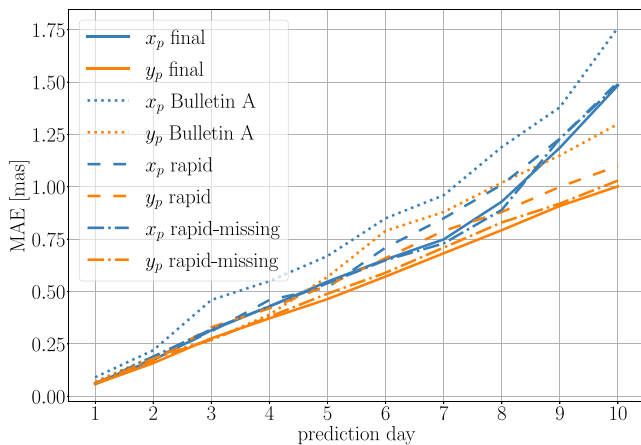
In the following section, we have analyzed the non-ideal cases according to Remark 6. These include the effect of missing data, using rapid EOP solutions, or IERS Bulletin A predictions.

#### 4.2.2. Simulation of Operational Predictions Under EOPCC Conditions

As mentioned in Remark 6, the EOPCC was done in an operational setting. Therefore, it is possible there were some problems with the availability and/or quality of data. As the final, most accurate EOPs data have a latency of up to several weeks, rapid EOPs data could be used to compensate for the need of having the latest values to predict the future. However, rapid EOPs data have some differences with respect to the final data. In addition, they have higher uncertainty compared to the final data. Therefore, using them to make predictions would result in predictions with larger uncertainty.

As of years 2021–2022, the difference between final and rapid EOPs data is mostly under 1 mas for the first few days. However, as IERS updates the rapid EOPs data continuously, the historical rapid EOPs data are no longer available on the server. Only since February 2021 have we saved the rapid EOPs data, meaning we only have data from early 2021. Therefore, we can analyze the differences between final and rapid EOPs data, model them with the Copula method, and attempt to simulate them for the study under the EOPCC conditions.

In addition, IERS provides Bulletin A predictions. This can also be used as a source of the latest values based on which to predict. Historical Bulletin A values are available during the EOPCC prediction period via the IERS webpage. It is important to note that both the rapid EOPs data and IERS Bulletin A have some missing data at certain times. During the study period of EOPCC, there is no missing IERS Bulletin A. However, we simulate



**Figure 10.** Analyzing the effect of using International Earth Rotation and Reference Systems Service (IERS) rapid earth orientation parameters data-with or without missing data and Bulletin A predictions for making predictions in the prediction interval of the first Earth Orientation Prediction Comparison Campaign. The results are shown on different days in terms of MAE (mas) for both the components  $x_p$  and  $y_p$ . Ideal, Bulletin A, rapid, and rapid-missing refer to, respectively, using IERS final, Bulletin A, rapid, and rapid data with some missing days.

the missing data for rapid EOPs data. Our assumptions for the simulation of rapid EOPs data are as follows:

- differences between rapid and final data under the study period of EOPCC have the same distribution as those in 2021–2022
- during 2021–2022 there were 17 days of missing IERS rapid EOPs data due to maintenance; we assume we also have a similar number days of missing rapid EOPs data during the EOPCC, for the study of missing values

While by no means these assumptions are fully accurate, we can get a picture, to some extent, of how the prediction performance would change. In any case, point number two is possibly a strong assumption and a big deterioration effect to most methods, as IERS normally provides data continuously without disruption. Having no rapid EOPs data would decrease the prediction accuracy, especially if these days are consecutive. Important to note is that the time of missing data in this simulation could be variable. We also analyze the effect of variability of the time of missing data in the simulations.

Under the mentioned conditions, Figure 10 presents one example of the effect of using rapid EOPs data-with or without missing data-and Bulletin A for making predictions, instead of final data. The location of the missing data is randomly chosen in the interval of prediction. In this figure, “ideal” means the case where we use the IERS final data to make predictions. In addition, rapid-missing refers to the case where we are using simulated rapid data which have some missing days.

From Figure 10 it could be understood that simulated rapid data perform almost equally well compared to the final data with a maximum of 11% reduction in prediction performance, while mostly below 4% reduction in prediction accuracy. It can be seen that the method is least affected when we use the simulated rapid EOPs data with missing values, as the method tries to learn how to interpolate in the hidden space. For the case of using simulated rapid data, this would result in, on average, a 3.49% and 11.80% reduction in prediction performance for  $x_p$  and  $y_p$ , respectively. The similar values for the reduction in prediction performance in the case of using simulated rapid data with missing values are 0.17% and 3.72%. In order to analyze the effect of the location of the missing days on the prediction performance, we used the extreme case of having all the missing days altogether. This resulted in a significant drop in the prediction performance of 26.12% and 37.82% for  $x_p$  and  $y_p$ , respectively. This confirms that consecutive days of missing values can be a major source of large prediction errors.

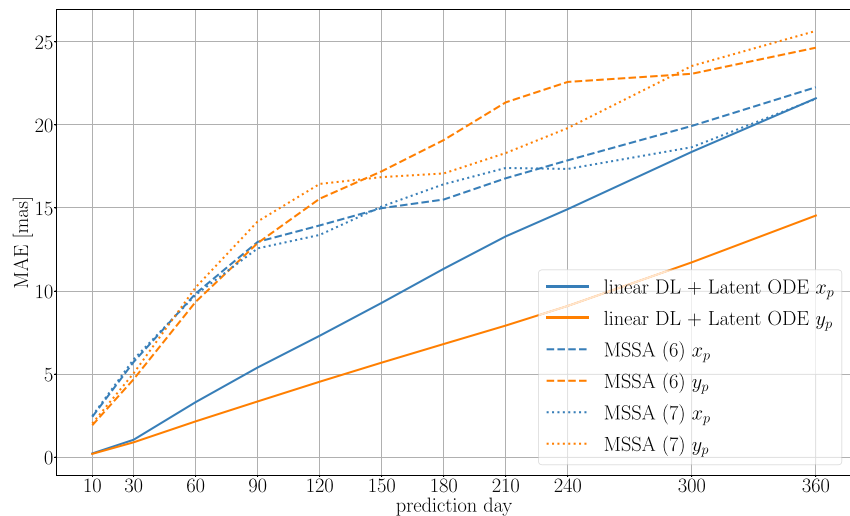
Not surprisingly, in the case of using Bulletin A to make predictions, a 28.87% and 21.66% reduction in prediction performance is expected for  $x_p$  and  $y_p$ , respectively. This is because the uncertainties in Bulletin A values, as stated in the files, are quite large, especially for longer prediction horizons. In addition, it can be shown that the prediction errors of Bulletin A are larger during the early years of providing the data. Therefore, during the years 2005–2008 this lower-quality predictions resulted in the reduction of prediction accuracy of the differential learning method. This would result in a relatively large drop in prediction accuracy. It is worth mentioning that even though non-linear models perform worse than the linear model used here, they are more robust when it comes to using data with higher uncertainty. Therefore, they have an advantage over the linear models in this case. Note, however, that in any case, our method is still much better, even when accounting for operational deficiencies.

### 4.3. Comparison With MSSA-Based Method

In order to properly compare the results of the proposed method with that of the MSSA-based study, the same time intervals as mentioned in Jin et al. (2021) are used for training and predicting based on the differential learning models. The properties of this study are as follows:

- prediction horizon of 360 days (long-term)
- time interval for training the model between 1 January 2000 to 6 January 2016





**Figure 11.** Comparison of prediction performance of the Multichannel Singular Spectrum Analysis-based method and linear differential learning (in combination with Latent ordinary differential equations) on different days in terms of MAE (mas) for both  $x_p$  and  $y_p$ .

- time interval for testing from 7 January 2016 to 31 December 2019, with window shifts of 28 days (new predictions made every 4 weeks)
- evaluation based on the IERS final EOPs data (IERS 14 C04)

It is important to note that the purpose of this comparison is to focus on the long-term prediction power of the differential learning method. As such, up to 360 days to the future are predicted and compared with predictions based on an MSSA-based study. First, like in the previous cases in order to find the best window length  $M$  for the analyses, the same analysis as mentioned in the two previous sections was performed. This reconfirmed the conclusion that the linear method in combination with Latent ODEs performs the best. The input sequence length, in this case, is, however, different from the studies before and assumes a value of  $M = 30$ . Similar to the previous case, in terms of MAE, for  $x_p$  and  $y_p$  components the linear method in combination with latent ODEs is at least 17.8% and 31.5% more accurate than the rest of the methods mentioned in Section 2.4.

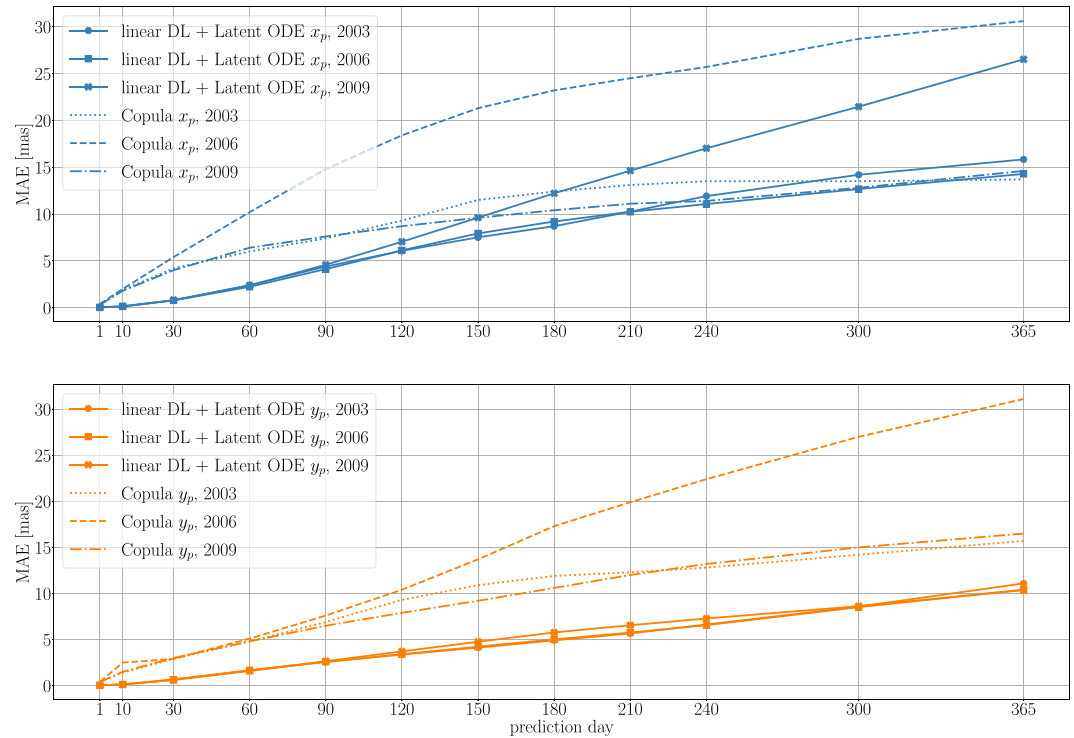
In Figure 11, the results of the proposed method are compared with those of the MSSA-based study. In this figure, the reconstruction degrees (number of eigenvalues used for reconstructing the time series from the trajectory matrix) are 6 and 7. The methods are therefore shown as MSSA(6) and MSSA(7).

In accordance with the results shown in Figure 11, the differential learning model tends to perform well for  $x_p$  across all days, but assumes more or less the same accuracy as that of MSSA(6) in the last few prediction days. In  $y_p$ , however, it outperforms both MSSA (6) and MSSA (7) by considerable margins. These are in alignment with the general better predictability of  $y_p$  compared to  $x_p$ , and the fact that the model is linear, which decreases rapidly in accuracy once the prediction horizon is longer. The average improvement across all days and both components is 54%, which is similar to the comparison with EOPCC in the previous section.

#### 4.4. Comparison With Copula-Based Method

Finally, the prediction accuracy of the proposed differential learning method is compared with the results obtained by using the Copula-based method, depicted in Figure 12 of Modiri et al. (2018). The conditions of the study are highlighted below:

- prediction horizon of 365 days (long-term)
- predictions analyzed in three different intervals 2003, 2006, and 2009
- training interval from the 1 January 1997 up to the last day of the year before the prediction year



**Figure 12.** Comparison between the MAE in mas of predictions of polar motion using Copula-based and linear differential learning methods (combined with Latent ordinary differential equations) in three different intervals.

- evaluation done on the IERS 08 C04 series (the data can be accessed via <https://hpiers.obspm.fr/eop-pc/index.php?index=C04%26lang=en>)

The comparison of the results based on both the Copula-based method (extracted from Modiri et al. [2018]; Figure 12) and the differential learning methods are given in Figure 12. More specifically, a similar study to the one discussed in Section 4.1 can be done for each of the three different prediction intervals mentioned above. Like the previous case  $M = 30$ , but in terms of MAE, for  $x_p$  and  $y_p$  components, the linear method in combination with Latent ODEs is at least 28.4% and 44.4% more accurate than the rest of the methods mentioned in Section 2.4.

Note that in Figure 12 like the previous cases we have used our linear method in combination with Latent ODEs. This is because this combination presented the best prediction performance on the validation set, compared to the other ones.

Based on the results presented in Figure 12, both the Copula-based and the differential learning methods have different performances in different intervals. The performance of the year 2006 deteriorated by a factor larger than 2 for the Copula-based method when compared to the results obtained by the same method in the interval 2003. This indicates that the methods are dependent on the characteristics of time series and the similarity between the training and the prediction intervals. Note, however, that the performance of  $y_p$  is consistently better in the differential learning method than in the Copula-based counterpart. For the very last day, however, the prediction performance of  $x_p$  of the differential learning method is worse in the years 2003 and 2009 than that of the Copula-based method. This demonstrates that the accuracy of the linear differential learning method is also dependent on the time interval during which it is evaluated. In the first few days of predictions (i.e., mostly short-term), the predictions are in some cases more than 80% more accurate. The average improvement across all days and both components is 52%.

**Table 2**

*Investigation of the Best Possible Combination of EAM Functions in Linear Models*

| EAM combination  | MSS (mas <sup>2</sup> ) |
|------------------|-------------------------|
| - (without)      | 4.81                    |
| AAM              | 3.78                    |
| OAM              | 1.86                    |
| HAM              | 2.08                    |
| SLAM             | 1.89                    |
| AAM,OAM          | 1.23                    |
| AAM,HAM          | 1.57                    |
| AAM,SLAM         | 1.63                    |
| OAM,HAM          | 1.74                    |
| OAM,SLAM         | 1.68                    |
| HAM,SLAM         | 1.93                    |
| AAM,OAM,HAM      | 1.19                    |
| AAM,OAM,SLAM     | 1.24                    |
| AAM,HAM,SLAM     | 1.55                    |
| OAM,HAM,SLAM     | 1.54                    |
| AAM,OAM,HAM,SLAM | 1.21                    |

*Note.* MSS is computed across the best possible combinations in which the model is linear.

## 5. Discussion

In this section, we intend to analyze the possible configurations of the differential learning method. In fact, there are several parameters in the algorithm that can determine the prediction performance in this method. In general, these can be categorized as follows:

- variations of differential learning method: linear, linear combined with Latent ODEs, Elman, Elman combined with Latent ODEs, LSTM, and LSTM combined with Latent ODEs,
- EAM input data: without, AAM, OAM, HAM, SLAM, and combinations thereof
- ways to generate pseudo-forecasts: addition of random noise, Copula simulation
- inclusion of uncertainties: with or without
- separate or combined models for  $x_p$  and  $y_p$
- different study conditions: under EOPPCC conditions, up to 2020, under MSSA-based method conditions, under Copula year 2003, under Copula year 2006, under Copula year 2009

As a result, by counting the number of all possible configurations, we would get 4,464 variations (In the case of no EAM, the possible ways of pseudo-forecasts are not relevant). In the following, we analyze the contribution of different parameters by analyzing the best performing configuration, while the parameter under consideration is constant and other parameters vary. In fact, since this analysis is similar to that of a high-dimensional optimization and analysis, it would be logical to analyze the effect of each parameter individually.

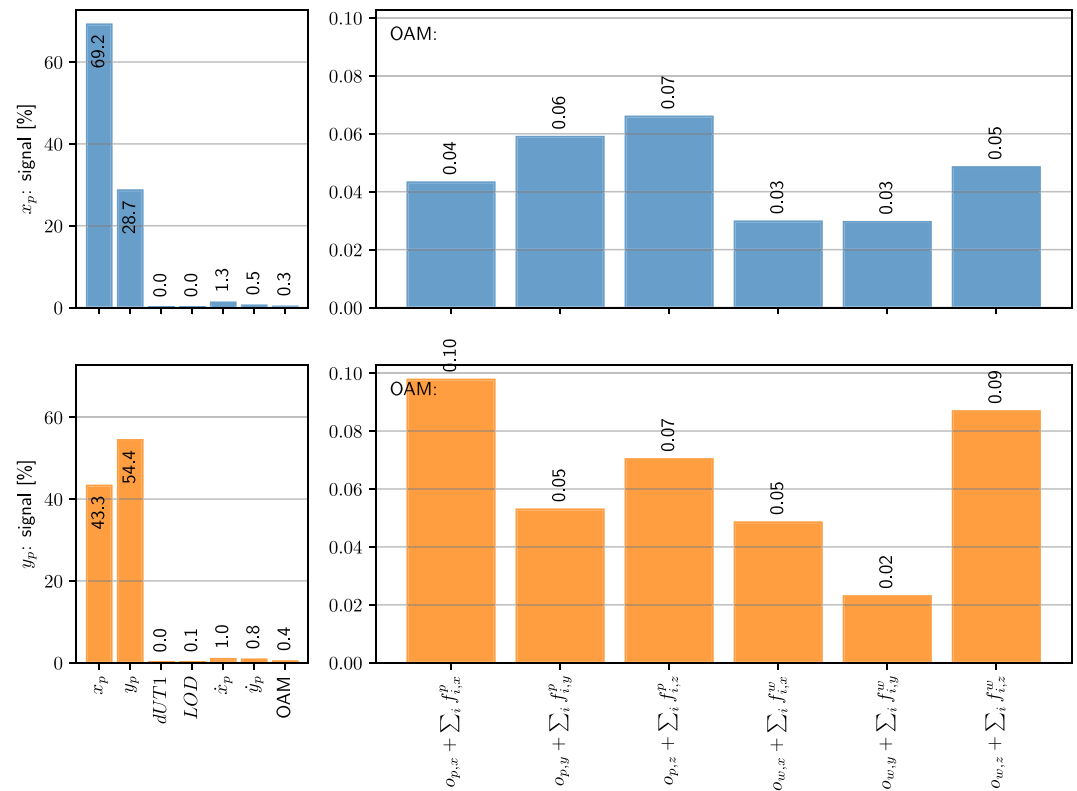
**Remark 7.** Note that we have already seen that the linear form of the differential learning method is performing the best for polar motion prediction. Therefore, in this section, this particular analysis is not performed. Note also that as mentioned before, using two separate models is better than having a model that simultaneously predicts both components  $x_p, y_p$ . We have observed in our analyses that this could have a significant impact on the prediction performance, particularly for the non-linear Elman model, when no EAM is used. As a rule of thumb, it can be said that using separate models for  $x_p$  and  $y_p$  is advantageous. Also, in terms of study, the EOPPCC conditions resulted in the most convincing results. Therefore, in the following, only the second, third, and fourth aforementioned analyses are presented.

### 5.1. Effect of Choice of EAM Combination

In order to decide which combination of EAM functions works the best for the prediction of polar motion components, we need to search across the 15 possible combinations. In addition, it is of interest to see the effect of the elimination of EAM data in the prediction problem. As an instance, we present the results for the linear model combined with Latent ODEs. Note that the criterion is the Mean Sum Squared (MSS, Equation 17, similar to Equation 16) of all MAE values in both  $x_p, y_p$ , and also across different days. The values of MSS of the best performing combination across linear models are mentioned in Table 2.

$$MSS_i = \frac{1}{N} \sum_{j=1}^N (O_{i,j} - P_{i,j})^2, \quad i = 1, \dots, h \quad (17)$$

From Table 2 it is inferred that the best combinations of EAM for linear models are [AAM, OAM, HAM] and [AAM, OAM, HAM, SLAM]. In fact, these two are close to each other and can be both used for the best performance. AAM per se does not result in a good performance. This attests to the fact that AAM, when used alone, is not a helpful feature to be used for the prediction of polar motion (Dill et al., 2021). Furthermore, combinations in which AAM is used alongside the OAM show better performance compared to the case of AAM alone, possibly due to the correlation between AAM and OAM. HAM and SLAM are less important than OAM. The results



**Figure 13.** Importance of input features to the model where only the oceanic angular momentum (OAM) is considered as the angular momentum function. The left parts show the importance of the seven features, namely,  $x_p$ ,  $y_p$ , dUT1, length of day,  $\dot{x}_p$ ,  $\dot{y}_p$ , and OAM. In the right parts, the importance of OAM is further divided into its components, that is,  $x$ ,  $y$ ,  $z$  (importance of observation and 6-day forecasts combined). Unlike Figure 7, the mass and motion terms are presented individually, denoted by  $p$  and  $w$  respectively. Like Figure 7, the upper figure is for prediction of  $x_p$ , while the lower figure is for the prediction of  $y_p$ .

with EAM in any case are better than when no EAM is used. This implies that EAM function help improves the prediction performance. It should be noted that for nonlinear LSTM and Elman models the results are mixed. Therefore no general conclusion regarding nonlinear models can be drawn.

An important point regarding the results in Table 2 is that the OAM functions are the most important ones for the prediction of polar motion and this is in agreement with the results of Dill et al. (2021). Dahlen (1976) discovered that oceans can increase the main period of polar motion by around 28 days. Many research works that followed showed that in fact, oceans are a major source of polar motion excitation (Bizouard & Seoane, 2010; Brzezinski & Nastula, 2002; Gross, 2008; Nastula & Ponte, 1999). Ocean processes like thermal expansion would also result in long-period changes in the polar motion (Landerer et al., 2009). Here we analyze the importance of mass and motion terms of the OAM functions on the prediction of polar motion (corresponding to the third case in Table 2). The results are shown in Figure 13. In this figure,  $p$  and  $w$  refer to the mass and motion terms, respectively. Note also that the importance of the 6-day forecasts and observations are summed up in order to show a clearer picture of which feature contributes the most to the prediction. Therefore, OAM represents six features here.

As it can be seen from Figure 13, it seems that the mass terms are more important than the motion terms for the prediction of both components. In  $x_p$ , mass term is approximately 54% more important than the motion term, while for  $y_p$ , the contribution of the mass term is approximately 37% more compared to the motion term.  $O_{p,y}$  is more important than  $O_{p,x}$  for the prediction of  $x_p$ . Similarly,  $O_{p,x}$  is more important than  $O_{p,y}$  for the prediction of  $y_p$ . This behavior can be explained by the Liouville equation of polar motion (Dill et al., 2020) because the changes in  $x_p$  are more related to the  $y$  component of excitation and vice versa. Thus the method has been able to imitate the behavior of the Liouville equation. The advantage of the differential learning method over the Liouville equation is its less reliance on EAM, as the Liouville equation is highly dependent on the quality of

**Table 3**  
*Investigation of the Effect of Approach to Generate the Pseudo-Forecasts on the Prediction Performance, in Different Studies*

| Study                                     | MSS of<br>addition of<br>random noise<br>approach<br>(mas <sup>2</sup> ) | MSS of<br>Copula<br>simulation<br>approach<br>(mas <sup>2</sup> ) |
|-------------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------|
| Baseline solution                         | 1.24                                                                     | 1.20                                                              |
| Under EOPPCC conditions                   | 1.28                                                                     | 1.19                                                              |
| Under MSSA-based method conditions        | 83.12                                                                    | 77.97                                                             |
| Under Copula-based method conditions 2003 | 210.33                                                                   | 158.00                                                            |
| Under Copula-based method conditions 2006 | 161.55                                                                   | 152.99                                                            |
| Under Copula-based method conditions 2009 | 387.54                                                                   | 328.80                                                            |

EAM data. In addition, differential learning can use an arbitrary number of features, while the Liouville equation is less capable of doing so.

## 5.2. Analyzing the Approach to Generate Pseudo-Forecasts

As mentioned in Section 3, there are two approaches to deal with the generation of pseudo-forecasts for the prediction problems: based on the addition of random noise, and modeling with Copula simulation and sampling. In order to analyze the effect of these approaches, the linear models (in combination with Latent ODEs) are considered. We have summarized the results in Table 3. In this table, different analyses are shown in the rows and the MSS of prediction performance using two different approaches in columns.

It can be inferred from Table 3 that the Copula simulation approach performs better than its random noise counterpart. This can be observed across all studies. Note, however, that in nonlinear models (Elman and LSTM) there is virtually no difference between the two approaches.

## 5.3. Effect of the Inclusion of Uncertainty on the Prediction Performance

As mentioned in Remark 3 uncertainties in the observations can be exploited for fitting a more reliable model. It can be shown that using uncertainties as weights in the loss function (see Supplementary Materials) can help improve the prediction performance. Uncertainties in observation files can provide valuable information regarding the quality of data. In order to assess the effect of the use of uncertainty in training the models, we consider here the linear models in combination with Latent ODEs. In this case, the method would be similar to using a weighted least squares problem (Magee, 1998). This is further discussed in the Supplementary Materials.

We have also done this procedure for nonlinear models based on the method presented by Kiani Shahvandi and Soja (2022). Note also that for nonlinear models in these studies sometimes the results are mixed, that is, no generalization can be made. Note, however, that except for these irregularities, to some very extent inclusion of uncertainty in data helps improve prediction performance.

In Table 4 the results based on the linear models in different studies are given. From this table, the positive impact of the inclusion of data uncertainty is demonstrated.

## 6. Conclusions

Based on the concept of learning the underlying differential equation describing the physical phenomenon, we introduced the differential learning algorithm for accurate time series prediction. We presented the algorithm in both linear form and based on recurrent neural networks, including Elmans and LSTMs. In a sensitivity analysis framework, we also presented the important factors for an arbitrary number of different features that can be used in the model. We determined a baseline solution in which we use historical JPL EOP data from 1976 to predict up to 2020. Furthermore, we presented three comparative studies, in which we thoroughly analyzed and compared the short- and long-term prediction performance of the proposed algorithms with the results of

**Table 4**  
*Investigation of the Effect of Using Uncertainty in the Observations on the Prediction Accuracy of Polar Motion Using Linear Differential Learning*

| Study                                     | MSS without uncertainty (mas <sup>2</sup> ) | MSS with uncertainty(mas <sup>2</sup> ) |
|-------------------------------------------|---------------------------------------------|-----------------------------------------|
| Baseline solution                         | 1.32                                        | 1.20                                    |
| Under EOPPCC conditions                   | 1.26                                        | 1.19                                    |
| Under MSSA-based method conditions        | 77.97                                       | 76.39                                   |
| Under Copula-based method conditions 2003 | 158.00                                      | 152.93                                  |
| Under Copula-based method conditions 2006 | 152.99                                      | 152.08                                  |
| Under Copula-based method conditions 2009 | 328.80                                      | 316.25                                  |

the best-performing method of the first EOPPC and also state-of-the-art results methods of multichannel and Copula-based singular spectrum analysis. We showed that the proposed method can improve the prediction results of the first EOPPC on average by 53%. The improvements for the first days can be as high as 86% and 82% for  $x_p$  and  $y_p$ , respectively. Simulation under the operational setting yielded the conclusion that the method still outperforms the best-performing model of the first EOPPC. In comparison to the MSSA-based method, our proposed method can improve the prediction on average by 54%, even though the predictions can be up to 88% more accurate for the first day of prediction. The similar improvement for the comparison with the Copula-based method is around 52% but can be as large as 92%. Compared to the aforementioned methods, our approach tends to outperform their prediction capability, thus setting a new reference baseline in terms of the short-term and long-term prediction accuracy of polar motion. In all the studies the linear model combined with Latent ODEs proved to be the best-performing model. Further improvement in prediction could be expected by using more accurate and effective angular momentum forecasts (Kiani Shahvandi et al., 2022).

As long as the prediction of polar motion is concerned, we recommend using linear differential learning models combined with Latent ODEs. However, the application of differential learning algorithms can be extended beyond the realm of Earth orientation parameters. In the field of geophysics, where most phenomena follow differential equations, it is of high importance to take their nature into account. In case there is no definite form for the differential equations or they have uncertainties, it is beneficial to use a stochastic differential equation, coupled with the concept of learning to learn the underlying relations from data. It is possible that in many applications nonlinear models would present better prediction performance compared to linear ones.

## Appendix A

### A1. Numerical Values of the Baseline Solution in 4.1

Here we have shown the numerical values of all the methods mentioned in Section 2.4, under the conditions of the baseline solution in Section 4.1, in Tables A1 and A2, for  $x_p$  and  $y_p$ , respectively.

**Table A1**

*MAE (mas) of Prediction of  $x_p$  Component of Polar Motion Using the Six Different Methods Mentioned in Section 2.4*

| Method              | Prediction day | MAE (mas) |      |      |      |      |      |      |       |       |       |
|---------------------|----------------|-----------|------|------|------|------|------|------|-------|-------|-------|
|                     |                | 1         | 2    | 3    | 4    | 5    | 6    | 7    | 8     | 9     | 10    |
| Linear              |                | 0.06      | 0.19 | 0.45 | 0.57 | 0.73 | 0.88 | 1.01 | 1.22  | 1.53  | 1.76  |
| linear + Latent ODE |                | 0.05      | 0.16 | 0.32 | 0.46 | 0.6  | 0.73 | 0.89 | 1.09  | 1.36  | 1.68  |
| Elman               |                | 0.15      | 0.97 | 2.22 | 3.91 | 5.65 | 7.18 | 9.77 | 12.83 | 16.66 | 19.46 |
| Elman + Latent ODE  |                | 0.14      | 0.42 | 1.12 | 1.99 | 3.01 | 4.50 | 5.12 | 6.23  | 7.79  | 8.58  |
| LSTM                |                | 0.09      | 0.38 | 0.59 | 0.77 | 0.92 | 1.23 | 1.67 | 2.25  | 3.59  | 5.41  |
| LSTM + Latent ODE   |                | 0.08      | 0.25 | 0.47 | 0.68 | 0.81 | 0.93 | 1.34 | 1.88  | 2.54  | 3.12  |

*Note.* The results are presented for the baseline solution in Section 4.1

**Table A2**

*MAE (mas) of Prediction of  $y_p$  Component of Polar Motion Using the Six Different Methods Mentioned in Section 2.4*

| Method              | Prediction day | MAE (mas) |      |      |      |      |      |      |       |       |       |
|---------------------|----------------|-----------|------|------|------|------|------|------|-------|-------|-------|
|                     |                | 1         | 2    | 3    | 4    | 5    | 6    | 7    | 8     | 9     | 10    |
| Linear              |                | 0.06      | 0.17 | 0.31 | 0.42 | 0.60 | 0.70 | 0.80 | 0.93  | 1.09  | 1.22  |
| linear + Latent ODE |                | 0.05      | 0.14 | 0.25 | 0.37 | 0.48 | 0.59 | 0.71 | 0.84  | 1.01  | 1.18  |
| Elman               |                | 0.13      | 0.76 | 1.89 | 2.88 | 4.10 | 6.83 | 8.91 | 12.01 | 15.44 | 19.25 |
| Elman + Latent ODE  |                | 0.11      | 0.37 | 0.92 | 1.71 | 2.86 | 4.18 | 5.00 | 6.03  | 7.59  | 8.23  |
| LSTM                |                | 0.08      | 0.33 | 0.46 | 0.69 | 0.88 | 1.14 | 1.47 | 2.09  | 2.92  | 3.54  |
| LSTM + Latent ODE   |                | 0.08      | 0.19 | 0.39 | 0.61 | 0.75 | 0.89 | 1.20 | 1.67  | 2.15  | 2.57  |

*Note.* The results are presented for the baseline solution in Section 4.1



## Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

## Data Availability Statement

IERS EOPs data can be accessed via <https://www.iers.org/IERS/EN/DataProducts/EarthOrientationData/eop.html>. IERS 08 C04 series used in the last study can also be accessed via <https://hpiers.obspm.fr/eop-pc/index.php?index=C04%26lang=en>. JPL EOPs data can be accessed at <https://keof.jpl.nasa.gov/combinations/2019/>. EAM data of GFZ German Research Center for Geosciences are available at <http://rz-vm115.gfz-potsdam.de:8080/repository>. The code and all the data used in the study are available alongside the manuscript at the repository <https://doi.org/10.5281/zenodo.7015400>.

## Acknowledgments

The authors would like to thank Dr. Grzegorz Kłopotek for his constructive comments on the original draft of the paper.

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